WORKSHEET

Chapter 9: Areas between curves (2)

1 Find the area enclosed by the functions $f(x) = 4x^2$ and $g(x) = x^2 + 3$.

2 Find the area enclosed by the functions $f(x) = x^2$ and $g(x) = x$.

3 Find the area enclosed by $y = x^3$ and $y = x$.

4 Find the area enclosed by $y = x^2$ and $y = \sqrt{x}$.

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5 Find the area enclosed by $y = 6 - x - 2x^2$ and $y = 4x + 3$ between $x = -4$ and $x = 2$.

6 Find the area enclosed by $y + x^2 = 6$ and $y + 2x - 3 = 0$.

7 Find the area enclosed by $y = -x^3 + 6x$ and $y = -x^2$.

8 Find the area enclosed by $y = \frac{-x^3}{3} + 2x^2$ and $y = -x^2 + 4x$.

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Answers

1 4

2 $\frac{1}{6}$

3 $\frac{1}{2}$

4 $\frac{1}{3}$

5 $28\frac{7}{12}$

6 $10\frac{2}{3}$

7 About 21.083

8 4

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